Netflix Movie Recommendation System

Project Report
Collaborative Filtering using SVD

Project Type	Machine Learning / Recommendation System
Primary Technique	Collaborative Filtering with SVD
Evaluation Metric	RMSE with 3-Fold Cross-Validation
Recommendation Output	Top 5 personalized movie recommendations
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This report is based on the uploaded Jupyter Notebook and documents the implemented data preparation, filtering, SVD-based recommendation model, evaluation, and personalized recommendation workflow.

1. Executive Summary

The project builds a personalized movie recommendation system from Netflix customer rating data. The raw rating data is transformed into Customer ID, Rating, and Movie ID. Lower-activity movies and customers are filtered using 60th-percentile rating-count benchmarks. An SVD collaborative filtering model from the Surprise library is evaluated using 3-fold cross-validation and RMSE. Finally, estimated ratings are generated for customer 1331154 and the five highest-scoring movie recommendations are returned.

2. Project Objectives

• Extract Movie IDs from the raw Netflix rating data.

• Remove rows where Rating is missing and convert Cust_Id to integer.

• Measure movie and customer rating activity.

• Filter lower-rated movies and less-active customers.

• Build an SVD collaborative filtering model.

• Evaluate the model using RMSE and 3-fold cross-validation.

• Generate personalized movie recommendations for a selected customer.

3. Dataset and Data Preparation

The notebook uses a Netflix ratings dataset and a separate movie-title CSV. The rating dataset contains Cust_Id, Rating, and Movie_Id after preprocessing. The movie-title file contains Movie_Id, Year, and Name.

Data Item	Notebook Result
Raw rating dataset rows	24,058,263
Rows after removing NaN ratings	24,053,764
Unique customers	470,758
Unique Movie IDs in rating data	4,499
Movie title records	17,770
Modeling columns	Cust_Id, Movie_Id, Rating

Data cleaning includes identifying movie boundaries from NaN ratings, creating Movie_Id, removing missing ratings, and converting Cust_Id to integer.

4. Pre-Filtering Strategy

The notebook calculates the number of ratings per movie and per customer. The benchmark is the 60th percentile of each rating-count distribution. Movies with fewer than 908 ratings and customers with fewer than 36 ratings are removed.

Filter	Benchmark	Removed	Result
Movies	908 ratings	2,699	1,800 movies remain

Customers	36 ratings	282,042	Filtered from 470,758
Rating records	Both filters applied	—	19,695,836 rows remain

5. Recommendation Model

The recommendation model uses Singular Value Decomposition (SVD) from the Surprise library. For quick runtime, the notebook prepares the first 100,000 rows of the filtered dataset for model evaluation. The input fields are Cust_Id, Movie_Id, and Rating.

6. Model Evaluation

The recorded 3-fold test RMSE values are 1.0147, 1.0146, and 1.0261. Their arithmetic mean is approximately 1.0185.

Fold	Test RMSE
1	1.0147
2	1.0146
3	1.0261
Mean	1.0185

The notebook's evaluation is based on the 100,000-row model dataset, so the reported RMSE should be interpreted in that context.

7. Personalized Recommendation

Customer 1331154 is selected for the recommendation demonstration. The customer has rated 253 unique movies. After removing excluded movies from the title data, 15,071 candidate records receive predicted scores. The candidates are sorted by estimated rating in descending order.

Rank	Movie	Estimated Score
1	Character	4.111528
2	Something's Gotta Give	3.913301
3	Lilo and Stitch	3.875684
4	Inspector Morse 31: Death Is Now My Neighbour	3.838021
5	What the #$*! Do We Know!?	3.778054

8. End-to-End Workflow

1. Load raw Netflix ratings.

2. Identify movie boundaries and create Movie_Id.

3. Remove missing ratings.

4. Convert Cust_Id to integer.

5. Calculate movie and customer rating counts.

6. Apply 60th-percentile filtering.

7. Load movie-title metadata.

8. Prepare Surprise dataset using 100,000 filtered rows.

9. Evaluate SVD with 3-fold cross-validation.

10. Predict scores for the selected customer.

11. Sort predictions and return the Top 5 movies.

9. Strengths and Limitations

Strengths

• Uses a standard collaborative filtering technique for user-rating data.

• Includes data cleaning and activity-based pre-filtering.

• Uses RMSE for quantitative model evaluation.

• Produces a clear personalized Top 5 recommendation output.

Limitations Identified in the Notebook

• Only the first 100,000 filtered rows are used for model evaluation for faster runtime.

• Only SVD is evaluated; alternative recommendation algorithms are not compared.

• The demonstration uses a fixed customer ID rather than a general user-input interface.

• The 60th-percentile filtering threshold is not compared with other thresholds.

• The notebook contains pandas SettingWithCopyWarning messages during two assignments.

10. Future Improvements

• Train and evaluate the final model on a larger or complete filtered dataset.

• Compare SVD with other collaborative filtering algorithms.

• Tune SVD hyperparameters and compare RMSE results.

• Create a reusable recommendation function for any valid customer ID.

• Build an interactive Streamlit or web interface.

• Add additional movie metadata for a hybrid recommendation approach.

• Evaluate ranking quality with suitable recommendation metrics.

11. Conclusion

The project demonstrates an end-to-end movie recommendation workflow, from preparing a large rating dataset through activity-based filtering and SVD collaborative filtering to personalized recommendation generation. The notebook records a mean 3-fold RMSE of approximately 1.0185 on its 100,000-row evaluation dataset and generates a ranked Top 5 list for the selected customer.

12. Technical Stack

Category	Technology
Programming	Python
Data Processing	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Recommendation	Scikit-Surprise, SVD
Environment	Jupyter Notebook / Google Colab workflow

Source: Uploaded notebook My_Netfilx_Recommendation_Project(1).ipynb